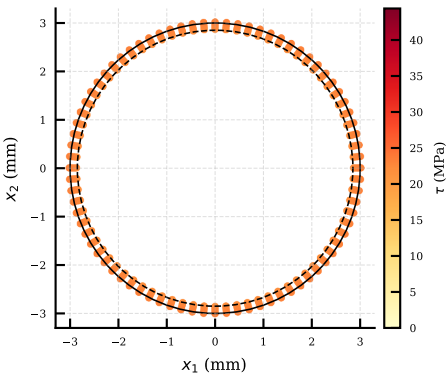
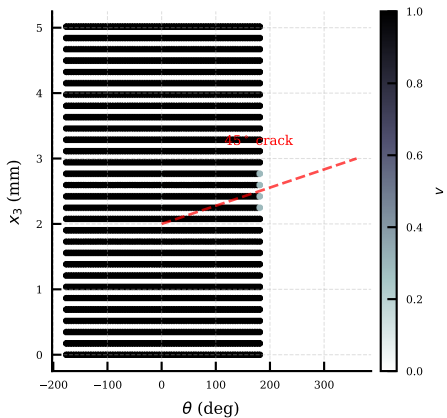


Torsion test at nucleation — cf. Kamarei et al. (2026) Fig. 8(c)

(a) τ at mid-section (step 33)



(b) Phase field v (unwrapped)



(c) 3D view with 45° crack

